CASE STUDY 2: AI Workflow Automation for Support Ticket Triage

Designing a Controlled, AI-Assisted Workflow Using Zapier for Scalable Support Operations

1. Problem Statement

The organization relies on manual, rule-based ticket triaging that depends heavily on individual judgment. This results in inconsistent prioritization, delayed handling of critical issues, and inefficient use of skilled support agents. The lack of real-time ticket intelligence limits leadership visibility into recurring issues and SLA risks.

2. Users & Stakeholders

Primary Users
• Support and operations teams managing high ticket volumes
• Support leads responsible for prioritization and escalation

Stakeholders
• Customer experience managers
• Business leadership focused on operational efficiency and service quality

3. Solution Options Considered

Option 1: Manual / Rule-Based Triage — Rejected
Limited contextual understanding, high operational effort, and poor scalability during peak demand.

Option 2: LLM-Only Automation — Rejected
High risk of misclassification, limited auditability, and lack of deterministic control over routing logic.

Option 3: Hybrid AI Automation with Zapier — Selected
A controlled automation workflow orchestrated using Zapier, combining NLP-based intent detection with business-defined prioritization rules. Generative AI is used selectively for ticket summarization and enrichment, while final routing decisions remain rule-driven and auditable.

4. High-Level Architecture & Workflow

1. Ticket ingestion from email, web forms, or CRM systems
2. Zapier triggers initiate the automation workflow
3. AI-assisted intent and urgency detection (advisory only)
4. Business rules engine determines final priority and routing
5. Human-in-the-loop review for high-risk or ambiguous cases
6. Zapier routes tickets to the appropriate support queue and tracks SLA status

5. Responsible AI Risks & Mitigation

Risk Area	Description	Potential Impact	Mitigation Strategy
Misclassification of Critical Tickets	AI may misinterpret ambiguous or poorly written tickets.	Delayed resolution, SLA breaches, customer dissatisfaction.	AI used in advisory mode; Zapier-enforced rules and human review for critical cases.
Over-Automation Without Context	Automation may ignore customer history or contractual priority.	Incorrect routing of premium or sensitive cases.	Metadata enrichment and rule-based overrides within Zapier workflows.

Bias in Intent Detection	Language patterns may skew AI interpretation.	Unequal service experience and fairness concerns.	Regular audits, diverse training data, and rule validation.
Lack of Transparency	Opaque AI decisions reduce trust.	Low adoption and compliance risk.	Zapier logs, rule-based routing, and explainable decision paths.

6. Expected Impact & Success Metrics

• Reduction in average ticket triage time
• Improved first-pass routing accuracy
• Higher SLA compliance for critical incidents
• Reduced manual intervention and re-routing
• Improved agent productivity and workload balance

7. Conclusion

This case study illustrates how Zapier can be used as an orchestration layer to operationalize AI responsibly. By combining AI-assisted intent detection with explicit business rules and human oversight, the solution achieves scalable automation while maintaining accountability, transparency, and trust.